

Lab experiments

Q1)

```
# Monthly Sales Data
```

```
month <- c("January", "February", "March", "April",  
"May")
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

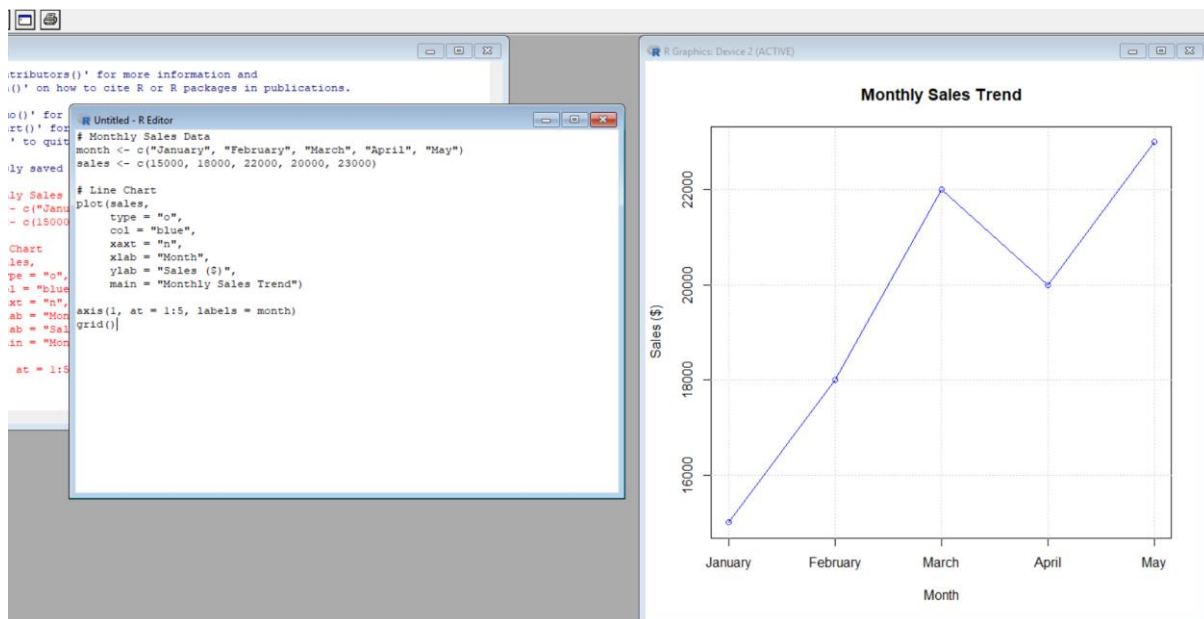
```
# Line Chart
```

```
plot(sales,  
     type = "o",  
     col = "blue",  
     xaxt = "n",  
     xlab = "Month",  
     ylab = "Sales ($)",  
     main = "Monthly Sales Trend")
```

```
axis(1, at = 1:5, labels = month)
```

```
grid()
```

Output:



Q2)

Product Sales Data

```
products <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
sales <- c(45000, 38000, 52000, 41000, 47000)
```

Bar Chart

```
barplot(sales,
```

```
      names.arg = products,
```

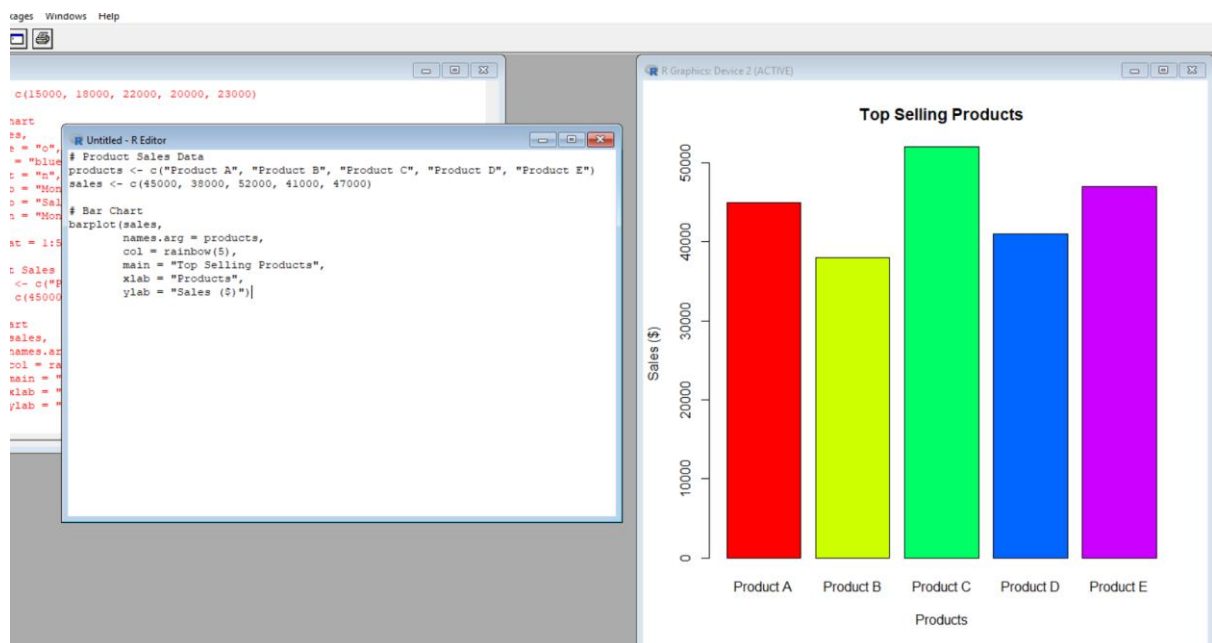
```
      col = rainbow(5),
```

```
      main = "Top Selling Products",
```

```
      xlab = "Products"
```

```
      ylab = "Sales ($)")
```

Outputs:



Q3)

Advertising Budget and Sales

```
budget <- c(2000, 2500, 3000, 2800, 3500)
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

Scatter Plot

```
plot(budget,
```

```
     sales,
```

```
     pch = 19,
```

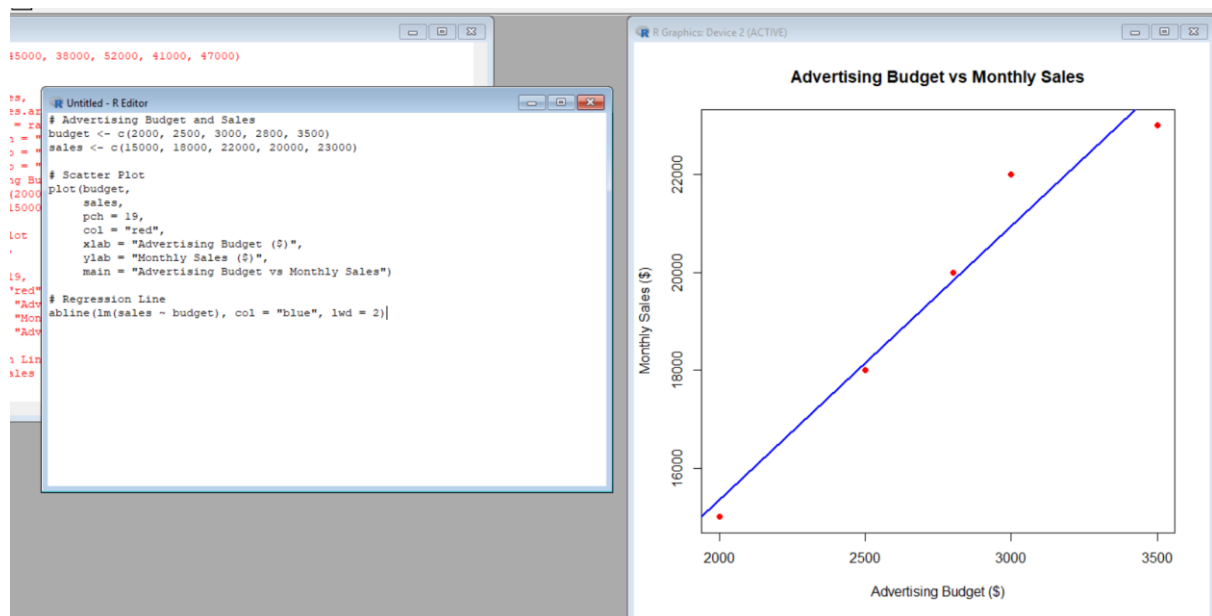
```
     col = "red",
```

```
     xlab = "Advertising Budget ($)",
```

```
     ylab = "Monthly Sales ($)"),
```

```
main = "Advertising Budget vs Monthly Sales")  
# Regression Line  
abline(lm(sales ~ budget), col = "blue", lwd = 2)
```

Output:



Q4)

```
# Display Two Charts Together
```

```
par(mfrow = c(1,2))
```

```
# Line Chart
```

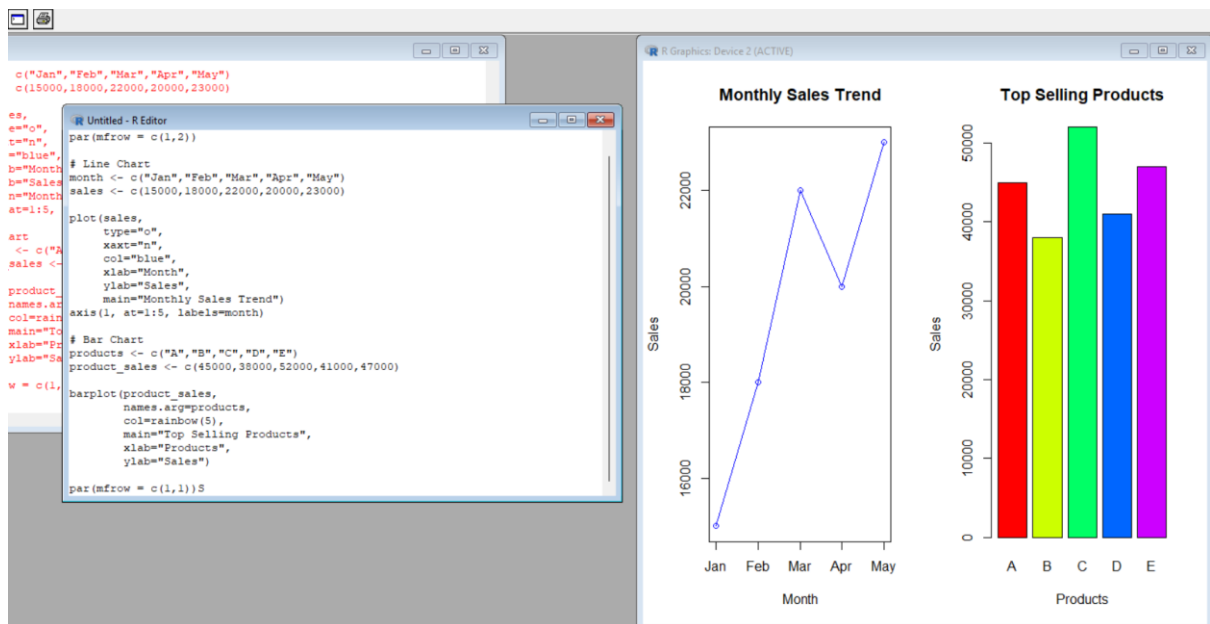
```
month <- c("Jan", "Feb", "Mar", "Apr", "May")
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
plot(sales,
```

```
type="o",
xaxt="n",
col="blue",
xlab="Month",
ylab="Sales",
main="Monthly Sales Trend")
axis(1, at=1:5, labels=month)
# Bar Chart
products <- c("A","B","C","D","E")
product_sales <-
c(45000,38000,52000,41000,47000)
barplot(product_sales,
        names.arg=products,
        col=rainbow(5),
        main="Top Selling Products",
        xlab="Products",
        ylab="Sales")
par(mfrow = c(1,1))
```

Output:



Q5)

Customer Age Data

age <- c(25, 30, 35, 28, 40)

Histogram

hist(age,

breaks = 5,

col = "lightgreen",

main = "Distribution of Customer Ages",

xlab = "Age",

ylab = "Frequency")

Output:

